

SUGUMAR K

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EXECUTIVE SUMMARY

Computer Science (IoT) undergraduate with hands-on experience in Java, Spring Boot, Machine Learning, and Full-Stack Development. Built AI-powered IoT security systems, deep learning healthcare applications, and enterprise web solutions. Seeking Software Development Engineer opportunities to apply strong problem-solving, backend development, and system design skills.

INTERNSHIP EXPERIENCE

Mist Solutions, Coimbatore

May 2025 – June 2025

Software Development Intern

- Engineered a PHP-based admin panel managing 500+ product records, streamlining order, sales, and inventory operations.
- Designed and implemented a business analytics dashboard to generate structured sales and order reports, reducing manual spreadsheet work by approximately 40%.
- Optimized MySQL database queries and backend workflows, reducing data retrieval time by approximately 30% and improving record accuracy.

TECHNICAL SKILLS

Programming: Java (Data Structures & Algorithms), Python, C++, SQL (MySQL)

Machine Learning & AI: Scikit-learn, TensorFlow, Keras, Deep Learning, Natural Language Processing (NLP), Ensemble Models, Pandas, NumPy, Matplotlib, Seaborn

Technologies / Frameworks: Spring Boot, JDBC, React.js, REST APIs, Flask, FastAPI

Tools & Platforms: GitHub, Jupyter Notebook, VS Code, Google Colab, Azure

PROJECTS

AI-Based Sandbox for Analyzing IoT Networks

Python | Flask | React | ESP32 | MQTT | Machine Learning

- Built a three-tier IoT security architecture integrating an ESP32 edge device, Flask backend, and React dashboard for real-time monitoring and control.
- Applied machine learning techniques for anomaly detection and attack classification, including DDoS, port scans, ICMP sweeps, and botnet traffic, using a hybrid XGBoost and Autoencoder model.
- Developed an MQTT-based feedback system to automatically send threat-response commands from the backend to ESP32 edge devices, enabling real-time mitigation of detected network attacks.

Metastatic Tissue Detection Using Deep Learning

Python | PyTorch | Computer Vision | FastAPI

- Built a deep learning-based histopathology image classification system using transfer learning with ResNet-34.
- Preprocessed and augmented 220,000+ medical images, achieving 95.2% Recall, 93.4% F1-score, and 0.96 ROC-AUC.
- Deployed a REST API using FastAPI for real-time predictions, enabling automated clinical screening support.

Multi-Agent RAG System for Verified Document Intelligence

Python | FastAPI | React | LangGraph | Qdrant | LLM | MinIO

- Developed a multi-agent Retrieval-Augmented Generation (RAG) system using LangGraph to orchestrate Retriever, Research, Verification, Correction, and Citation agents for reliable document-grounded question answering.
- Implemented hybrid BM25 and Qdrant vector search to improve document retrieval and reduce hallucinations.
- Built an automated verification and correction pipeline with structured citations and confidence scoring.

EDUCATION

Sri Krishna College of Technology, Coimbatore

B.E. Computer Science and Engineering (IoT)

Expected April 2027

CGPA: 7.8 / 10 (as of 6th semester)

Government Higher Secondary School, Eriyur

Higher Secondary Certificate (HSC)

May 2023

81%

Government High School, Pudhubalasamudiram

Secondary School Leaving Certificate (SSLC)

March 2021

CERTIFICATIONS

- Java Foundation Certification – Infosys
- Java OOPs Concepts – Infosys
- Spring Data JPA with Boot – Infosys
- Azure Fundamentals – Microsoft